

a long-range telescope in Leh, a radar in India's northeast region, and a multi-object tracking radar (MOTR) in Sriharikota. In addition, there are optical-infrared observatories at Bengaluru, Pune, Mt Abu, Nainital, and Hanle.

ISRO also collaborates with countries like United States, France, Australia, and Japan for exchange of information and data. ISRO's Project NETRA (Network for Space Object

Tracking and Analysis) tracks satellites in low earth orbits to prevent being hit by space debris or space weather incidents. NETRA will eventually become part of the global system for tracking, warning, and mitigating space debris. Also, several new facilities—including radars, telescopes, data processing centres, and a control centre—to spot, track, and catalogue objects as close as 10cm and as far away as 3400km, are being established.

Role of ISRO and DRDO

ISRO was set up in 1969 for realising the scientific and commercial potential of space, scrupulously avoiding space militarisation. Only in the last few years, ISRO has been collaborating with defence agencies to meet their specific requirements. For example, the current satellite programmes include battlefield communication and cross-border surveillance.

ISRO's space based assets include a fleet of fifty functional satellites, including communication satellites GSATs and INSATs, surveillance satellites CARTOSAT 2 and RE-SOURCESAT, and weather monitoring satellites. ISRO's remote sensing programme (IRS) consists of the largest constellation of remote sensing operational satellites in the world. Eleven operational satellites placed



Fig. 3: A visual representation of the SSA concept (Source: <https://www.esa.int>)

in polar Sun-synchronous orbit provide data in a variety of spatial, spectral, and temporal resolutions for various applications on ground.

The RISAT series of satellites are all-weather earth observation satellites, including synthetic aperture radar satellites RISAT-1 and RISAT-2 that have day and night viewing capabilities. The Indian Regional Navigation Satellite System (IRNSS constellation) is an autonomous regional satellite navigation system that provides accurate real-time positioning and timing services,



Fig. 4: A multiple object tracking radar (Source: <https://www.army.mil>)

covering a region extending 1500km around the border.

With the success of the geosynchronous satellite launch vehicle (GSLV-D5), ISRO is ready to take on satellites weighing in excess of two tons, an important prerequisite to the deployment of any weapons system. ISRO launched dedicated military satellites GSAT-6 and GSAT-7A in August 2015 and December 2018, respectively, for providing secure communication for the Armed Forces. In April 2019, ISRO

launched EMISAT (Electro-Magnetic Intelligence Satellite) that was jointly developed by ISRO and DRDO to strengthen the capability to intercept enemy radars.

The DRDO, with a network of 52 laboratories, was established in 1958 for developing defence technologies covering various fields, including missiles. Its Integrated Space Cell (ISC), operated jointly by the three services and DRDO, was set up in 2008 as part of programme to develop military space capabilities. Prithvi missiles, a range of short-range ballistic missiles, were developed in the 1980s under the Integrated Guided Missile Development Programme (IGMDP).

The Agni ballistic missiles, a range of MRBMs, IRBMs, and ICBMs developed by DRDO include Agni-V, a long-range deterrence ICBM with a range 6000km. Agni-V can carry multiple independently targetable re-entry vehicle payloads to provide counter-measures against anti-ballistic missile systems. DRDO is also developing an ICBM, codenamed Surya, which is a three-stage missile with solid and liquid fuel as propellant.

In 2006, DRDO unveiled an anti-ballistic missile (ABM) project, which has two ICBMs—the AAD (Advanced Air Defence) and PAD (Prithvi Air Defence) missiles. The